PAYROLL PRO

ENTERPRISE DESKTOP & CLOUD SUITE

OFFICIAL OPERATIONAL DOCUMENTATION

Instruction Manual & Statutory Reference Guide

A comprehensive guide covering employee onboarding, monthly gross-to-net payroll computation, loan amortizations, payslip customization, corporate bank schedules, and statutory compliance under Nigerian tax laws.

DOCUMENT VERSION Release 1.0.2 (Production Stable)	PRIMARY AUDIENCE Payroll Officers, HR Managers, Controllers
STATUTORY JURISDICTION Federal Republic of Nigeria (PITA / PRA 2014 / 2026 Tax Bill)	SYSTEM ARCHITECTURE Electron Native Desktop + Supabase Realtime Cloud

CONFIDENTIAL & PROPRIETARY — FOR AUTHORIZED COMPANY USE ONLY PAYROLL PRO ENTERPRISE © 2026

TOC Table of Contents

SECTION	MODULE & OPERATIONAL TOPIC	TARGET AUDIENCE
Section 1	**Executive Overview & System Architecture** Offline-first caching, Supabase Realtime sync, system specifications, and global navigation.	All Operators
Section 2	**Authentication & Multi-Company Workspaces** Cloud connection, Windows Account Picker, workspace PINs, and security question recovery.	Admins & Staff
Section 3	**Employee Directory & Staff Management** Quick-Add spreadsheet bulk entry, Standard Profile setup, Department select dropdown, and 360° modal.	HR & Payroll
Section 4	**Monthly Payroll Computation & Finalization** Gross-to-net computation, stint prorations, monthly overrides, period snapshot locking, and NUBAN bank schedules.	Payroll & Finance
Section 5	**Loans & Salary Advances Ledger** Disbursing facilities, auto-deduction integration, pause/resume, and manual repayment entries.	Finance & Accounts
Section 6	**WYSIWYG Payslip Designer & Batch Export** Layout presets (Modern, Classic, Compact), logo branding, cryptographic QR code verification, and PDF printing.	HR & Finance
Section 7	**Statutory Compliance & Mathematical Models** PITA 2011 & 2026 Tax Bill progressive bands, CRA formulas, PRA 2014 BHT 8%/10% Pension, and worked examples.	Compliance & Audit
Section 8	**Role-Based Access Control, Presence & Security Auditing** Admin vs Officer vs Auditor matrix, live operator presence, instant deactivation, and immutable audit logs.	System Admins
Section 9	**Troubleshooting, Data Backup & Disaster Recovery** Encrypted JSON backups, silent background sync, common edge-cases, and recovery workflows.	Admins & IT

Global Keyboard Shortcuts

Payroll Pro provides ergonomic keyboard triggers across all active workspaces:

SHORTCUT KEY	DESTINATION / ACTION	DESCRIPTION
`Alt + 1`	**Executive Dashboard**	Navigate directly to payroll KPIs, total statutory liabilities, and department breakdown.
`Alt + 2`	**Staff Directory**	Open employee directory ledger, search filters, and bulk onboarding tools.
`Alt + 3`	**Payroll Processing**	Jump to current period gross-to-net computation ledger and bank exports.
`Alt + 4`	**Loans & Advances**	Open loan register, repayments, and facility disbursement form.

SHORTCUT KEY	DESTINATION / ACTION	DESCRIPTION
`Alt + 5`	**Payslip Designer**	Open visual WYSIWYG payslip customizer and layout templates.
`Alt + 6`	**Company Settings**	Manage company profile, tax rules, departments, backups, and audit trails.
`Escape (Esc)`	**Dismiss / Close**	Immediately dismiss any open modal, dialog, or floating action menu.

01 Executive Overview & System Architecture

Payroll Pro is a specialized enterprise application built to handle the complex statutory, tax, and pension realities of operating in Nigeria. It eliminates spreadsheet errors, enforces compliance with federal and state regulations, and streamlines disbursement through Nigerian commercial banks.

🔒 Dual-Mode Multi-Tenancy

Operates fully offline on isolated local storage or connects seamlessly to private Supabase PostgreSQL instances for multi-branch team collaboration.

⚡ Zero-Latency Role Enforcement

Role promotions, demotions, and operator deactivations take effect across remote operator windows in sub-seconds via Supabase Realtime CDC.

NG Complete Nigerian Compliance

Built-in models for PITA 2011, Finance Act 2020, Pension Reform Act 2014, and the upcoming 2026 Nigerian Tax Reform Bill progressive brackets.

🛡️ Non-Destructive Background Sync

Silent multi-layered polling and WebSocket updates refresh data in memory without table unmounting, layout shifting, or screen flickering.

System Architecture & Data Flow

Offline-First Resilience: All changes made by operators are immediately written to local memory and persistent storage, followed by optimistic background synchronization to Supabase Cloud. If network connectivity drops, operators can continue processing payroll without interruption; changes sync automatically upon reconnection.

COMPONENT	TECHNOLOGY	OPERATIONAL FUNCTION
Desktop Shell	Electron 44 (x64)	Provides high-performance native Windows integration, tray support, local file system access, and native print-to-PDF generation.
User Interface	React 18 & Tailwind CSS	Reactive, dark-mode ready UI with zero third-party component bloat, tailored for high-volume tabular accounting operations.
Computation Engine	Pure TypeScript	Deterministic, audit-ready mathematical engine implementing Nigerian statutory emoluments, CRA formulas, and marginal tax bands.
Cloud Database	Supabase (PostgreSQL 14+)	Multi-tenant relational database with foreign key cascades, Row Level Security (RLS), and Realtime WebSocket replication.
Export Engine	SheetJS & xlsx-js-style	Generates styled Excel corporate bank schedules (NUBAN format) and printable A4 payslips.

02 Authentication & Multi-Company Workspaces

Payroll Pro isolates data by legal corporate entities. Operators sign into the system with their verified credentials and then select or unlock their target company workspace.

2.1 Cloud Staff Access Gate

Upon launching the application, the system presents the Cloud & Staff Access Gate:

1. **Enter Operator Credentials:** Enter your registered Work Email or Username and Password.
2. **Automatic Session Retention:** Check *"Remember me on this workstation"* to bypass subsequent login prompts on secure company computers.
3. **First-Time Admin Setup:** If connecting to a newly deployed database, the gate prompts you to register the Master Administrator with an initial security question and answer.
4. **Connecting a Private Server:** Click **Connect Server** in the top-right corner to input your organization's private Supabase URL and Anon Public Key.

2.2 Windows OS-Style Account Picker

Once authenticated, the system presents the interactive workspace gallery:

> **Multi-Entity Switching:** Holding companies and accounting firms managing multiple subsidiaries can switch between workspaces in seconds. Each company maintains its own distinct logo, password, staff register, and locked historical snapshots.

WORKSPACE FEATURE	DESCRIPTION & OPERATIONAL RULE
Company Profile Card	Displays the company's legal name, logo, active staff count, and lock status.
Workspace PIN / Password	If a workspace has password protection enabled, clicking the card prompts for the confidential PIN/password before granting entry.
+ Add Company Workspace	Administrators can create a new entity by providing the legal business name, RC/CAC number, registered address, and default state of residence.
Authorized Team Members	Clicking **Team Members** in the top ribbon opens the central operator delegation ledger where roles and presence are administered.

2.3 Self-Service Password Recovery

If an operator forgets their login credentials, they can perform self-service recovery:

1. Click **Forgot Password?** on the login screen.
2. Enter the registered **Work Email or Username**.
3. The system verifies the account and presents the confidential **Security Question** registered during onboarding.
4. Provide the exact **Security Answer** to immediately unlock the account and set a fresh confidential password.

03 Employee Directory & Staff Management

The Employee Directory is the operational center for personnel records, salary contracts, statutory tax and pension settings, and mid-month tenure stints.

3.1 Quick-Add Spreadsheet Form (Bulk Onboarding)

Designed specifically for rapidly populating staff registers during new company setup or bulk seasonal hiring.

> **Keyboard Power Workflow:** Pressing **Enter** or **Tab** at the end of a row automatically validates the current staff member, commits the entry, and creates a fresh empty row with the cursor placed in the Full Name field.

COLUMN	INPUT TYPE	RULES & VALIDATION
Full Name	Text	Required. Automatically parses First Name and Last Name.
Monthly Salary (₦)	Number / Currency	Monthly Gross. Supports arbitrary decimal precision and empty states while typing.
Employment Type	Select Dropdown	Select from Full-Time, Contract, Intern / NYSC, or Part-Time.
Department	Native Select Dropdown	Select from company operational units (Operations, Finance, Sales, Admin, IT).

3.2 Standard Employee Profile Modal

Used when detailed statutory registration or individual contract terms are required.

👤 Identity & Role

Staff ID Code, First Name, Last Name, native Department select dropdown, Job Title, Work Email, and Phone.

💰 Compensation & Rent

Base Monthly Gross (₦), Annual Rent Emolument (for statutory Rent Relief), and monthly salary overrides.

🏛️ Statutory Tax & Pension

Subject to Pension (8%), Subject to PAYE, licensed PFA selection, 11-digit RSA Number, Taxpayer ID, and State of Residence.

🏦 Bank & Disbursement

Official Bank Name (from licensed Nigerian banking directory) and validated 10-digit NUBAN account number.

3.3 Employee 360° Comprehensive View

Clicking any employee in the directory launches a balanced, high-fidelity **50/50 two-column landscape dashboard**:

- **Left Column (Identity & Profile):** Staff avatar, contact cards with one-click copy, statutory registration numbers, and banking details.
- **Right Column (Live Payroll Cycle & Emoluments):** Real-time month picker displaying exact emolument splits (Basic, Housing, Transport), CRA relief, PAYE tax, and net take-home pay for that specific cycle.
- **Edit Launchpad:** One-click button to jump directly into full profile editing.

04 Monthly Payroll Computation & Finalization

The Payroll Processing module computes gross-to-net calculations in real-time, factoring in mid-month stint prorations, monthly bonuses, loan repayments, and statutory tax deductions.

4.1 Computation Workflow

1. **Select Cycle:** Click the target month (e.g., *October*) and year from the top period ribbon.
2. **Review Prorations:** Staff joining or leaving mid-month have their gross earnings automatically prorated by active calendar days.
3. **Apply Active Loan Deductions:** Scheduled staff loan and advance deductions are automatically integrated into each employee's deductions column.
4. **Audit Ledger Figures:** Inspect Gross Pay, CRA Relief, Taxable Income, PAYE Tax, Pension (8%), Employer Pension (10%), and Net Take-Home Pay.
5. **Lock Period Snapshot:** Finalize the run to archive the numbers permanently against accidental retrospective modification.
6. **Generate Bank Schedule:** Export the validated NUBAN schedule for corporate internet banking upload.

4.2 Stint & Proration Logic

When an employee has a recorded Start Date or Exit Date falling within the active processing month, Payroll Pro applies exact calendar proration:

```
// Calendar Day Tenure Proration Formula
Proration Factor = Active Days in Month / Total Calendar Days in Month
Prorated Gross = Base Gross Salary * Proration Factor
```

4.3 Period Snapshot Locking & Historical Archiving

To protect accounting integrity, completed payroll runs must be locked:

SNAPSHOT PARAMETER	COMPLIANCE & OPERATIONAL RULE
Lock Confirmation	Presents an executive summary: Total Gross, Total PAYE, Total Pension, and Net Disbursed.
Audit Attribution	Records the authorizing operator's name, role, timestamp, and optional run notes.
Immutable Lock State	Once locked, the period displays a golden padlock (🔒 `Locked`). Subsequent changes to employee base salaries in future months will not alter this finalized historical ledger.
Unlock Privilege	Only Master Administrators can unlock a finalized run, and doing so generates an immediate priority entry in the system audit log.

4.4 Corporate Bank Schedule Generation (NUBAN)

Clicking **Export Bank Schedule** generates an Excel workbook (`.xlsx`) formatted to CBN electronic payment standards, pre-formatted for direct upload into:

🏦 **AccessOnline / Access Bank**

🏦 **GTBank GAPS**

Standard corporate batch disbursement template with 10-digit NUBAN and sort codes.

Guaranty Trust Automated Payment System corporate batch upload schedule.

🏦 Zenith Corporate iBank

Zenith Bank electronic salary disbursement format with automated narration strings.

🏦 UBA BankCollect / FirstBank

Compatible with United Bank for Africa and FirstBank FirstPay corporate salary portals.

05 Loans & Salary Advances Ledger

Manage company lending programs, short-term salary advances, monthly payroll amortizations, and balance tracking without external spreadsheets.

5.1 Facility Types

FACILITY TYPE	STANDARD TERM	OPERATIONAL PURPOSE & USAGE
Salary Advance	1 to 3 Months	Emergency mid-month advance deducted from upcoming salary cycles.
Staff Loan	6 to 24 Months	General personal loan amortized evenly across monthly payrolls.
Vehicle / Car Loan	12 to 48 Months	Automobile financing with scheduled monthly principal deductions.
Housing Advance	6 to 12 Months	Annual accommodation support facility amortized over the lease term.
Emergency Loan	1 to 6 Months	Compassionate relief facility with flexible repayment scheduling.

5.2 Disbursing a Loan Facility

1. Navigate to **Loans & Advances** (`Alt + 4`).
2. Click **+ New Facility**.
3. Select the **Employee** from the searchable dropdown.
4. Select the **Facility Type** and enter the **Principal Amount (₦)**.
5. Enter the **Monthly Deduction Amount (₦)** (or loan duration in months).
6. Select the **Repayment Start Period** (e.g., `2026-may`).
7. Click **Create Facility**. The monthly deduction automatically binds to payroll processing starting from the chosen cycle.

5.3 Auto-Deduction & Lifecycle Tracking

> **Automatic Completion:** When a monthly payroll run is executed, active loan deductions are deducted from the employee's net pay. The facility's remaining balance decreases automatically. When the balance hits **₦0.00**, the facility status flips to **Completed** and deductions automatically cease.

5.4 Facility Actions: Pause, Resume & Manual Payments

Clicking the minimalist 3-dot action menu (•••) on any facility provides:

- **Pause Facility:** Temporarily halts payroll deductions during staff hardship or special leave.
- **Resume Facility:** Restores monthly payroll amortizations.
- **Record Manual Payment:** If an employee repays cash directly, record the transaction to reduce the balance and log the receipt in the loan's audit history.

- **Edit Facility:** Adjust monthly deduction amounts, remaining balance, or facility terms.

06 WYSIWYG Payslip Designer & Batch Export

Design and distribute branded, tamper-evident payslips featuring corporate logos, custom color schemes, authorized digital signatures, and cryptographic QR verification.

6.1 Layout Templates

🎨 Modern Template

Clean, borderless card structure with vibrant brand accents, bold net pay callouts, and modern typography.

📋 Classic Corporate

Traditional accounting ledger styling with formal table borders, comprehensive statutory disclosures, and signature lines.

📄 Compact Format

Space-efficient layout designed for high-volume printing, thermal slips, or fitting two payslips per standard A4 sheet.

📑 Executive Statement

Comprehensive layout with year-to-date (YTD) cumulative earnings, pension history, and tax deduction tracking.

6.2 Branding, Watermarks & Cryptographic QR Codes

SECURITY / DESIGN FEATURE	OPERATIONAL BENEFIT & IMPLEMENTATION
High-Res Company Logo	Upload PNG/SVG logos for sharp rendering on both physical paper and digital PDF files.
Authorized Signature Stamp	Include official digital signature stamps and date lines for immediate bank/visa verification validity.
Cryptographic QR Code	Generates a tamper-evident QR code embedding the staff ID, month key, and hash signature. Scanning verifies authenticity and combats forged payslips.
Confidential Watermark	Subtle background watermark discouraging unauthorized duplication or external distribution.

6.3 Single & Batch PDF Printing

1. **Single Slip:** In the Payroll computation ledger, click **Print Payslip** on any employee's row.
2. **Batch Slips:** Click **Batch Print / PDF** in the payroll action ribbon to render an aggregated print-ready file. Each staff member's payslip is cleanly isolated with CSS page-break rules for standard A4 printing.

07 Statutory Compliance & Mathematical Models

Payroll Pro implements the exact statutory calculation rules of the **Personal Income Tax Act (PITA 2011)**, the **Pension Reform Act 2014 (PRA 2014)**, and the **2026 Nigerian Tax Reform Bill**.

7.1 Emolument Allocations & BHT Base

Gross earnings are automatically allocated into statutory allowances:

- **Basic Salary:** 40% of Gross (Pensionable & Taxable)
- **Housing Allowance:** 30% of Gross (Pensionable & Taxable)
- **Transport Allowance:** 20% of Gross (Pensionable & Taxable)
- **Other Allowances:** 10% of Gross (Non-Pensionable & Taxable)

The sum forms the statutory **BHT Emolument**: `BHT = Basic + Housing + Transport = 90% of Gross`.

7.2 Statutory Pension (PRA 2014)

```
Employee Pension Deduction = BHT * 8% (7.2% of Gross — 100% Tax Deductible)
Employer Pension Contribution = BHT * 10% (9.0% of Gross — Company Expense)
```

7.3 Consolidated Relief Allowance (CRA)

STATUTORY REGIME	ANNUAL CRA FORMULA	MONTHLY EQUIVALENT
PITA 2011 / Finance Act	`max(₦200,000, 1% Gross) + 20% Gross`	`max(₦16,666.67, 1% Gross) + 20% Gross`
2026 Tax Reform Bill	`₦800,000 Base + 20% Gross`	`₦66,666.67 Base + 20% Gross`

7.4 Progressive Tax Brackets (PITA 2011 vs 2026 Reform)

PITA 2011 BAND (ANNUAL)	RATE	2026 TAX REFORM BILL BAND (ANNUAL)	RATE
First ₦300,000 (₦25,000/mo)	**7%**	First ₦2,200,000 (₦183,333.33/mo)	**15%**
Next ₦300,000 (₦25,000/mo)	**11%**	Next ₦9,000,000 (₦750,000.00/mo)	**18%**
Next ₦500,000 (₦41,666.67/mo)	**15%**	Next ₦13,000,000 (₦1,083,333.33/mo)	**21%**
Next ₦500,000 (₦41,666.67/mo)	**19%**	Next ₦25,000,000 (₦2,083,333.33/mo)	**23%**
Next ₦1,600,000 (₦133,333.33/mo)	**21%**	Above ₦49,200,000 (₦4,100,000.00/mo)	**25%**
Above ₦3,200,000 (₦266,666.67/mo)	**24%**	—	—

7.5 Worked Numerical Example (₦500,000 Monthly Gross)

```
1. Monthly Gross: ₦500,000.00 (Annual: ₦6,000,000.00)
2. BHT Emolument: ₦450,000.00 (Basic ₦200k + Housing ₦150k + Transport ₦100k)
3. Employee Pension (8%): ₦36,000.00 (Tax Deductible)
```

4. Employer Pension (10%): ₦45,000.00 (Company Liability)
5. Monthly CRA Relief: ₦116,666.67 (₦200k/12 + 20% of ₦500k)
6. Chargeable Taxable Income: ₦500,000 - ₦36,000 - ₦116,666.67 = ₦347,333.33
7. Monthly PAYE Tax (PITA): ₦66,026.67
8. Net Take-Home Pay: ₦500,000 - ₦36,000 - ₦66,026.67 = ₦397,973.33

08 Role-Based Access Control, Presence & Security Auditing

Payroll Pro enforces separation of duties through a three-tier Role-Based Access Control (RBAC) architecture.

8.1 Operator Access Matrix

OPERATIONAL CAPABILITY	ADMINISTRATOR	PAYROLL OFFICER	AUDITOR
Configure Cloud Server & Database Keys	✅ Full	❌ Denied	❌ Denied
Manage Operators, PINs & Roles	✅ Full	❌ Denied	❌ Denied
Add, Edit & Deactivate Staff Records	✅ Full	✅ Full	👁 Read-Only
Run Monthly Payroll Calculations	✅ Full	✅ Full	👁 Read-Only
Lock & Finalize Historical Runs	✅ Full	✅ Full	❌ Denied
Export Bank NUBAN Schedules	✅ Full	✅ Full	👁 Audit Copy
View System Security Audit Trail	✅ Full	❌ Denied	👁 Read-Only
Modify Tax Variables & CRA Formulas	✅ Full	❌ Denied	❌ Denied

8.2 Live Operator Presence & Real-Time Tracking

When authorized team members open Payroll Pro across different locations, WebSocket presence tracking broadcasts active sessions. The team ledger displays live green indicator pills (`● Online since 09:14 AM`) including the company workspace they are currently viewing.

8.3 Zero-Latency Forced Deactivation

> **Security Enforcement:** If an employee resigns or a security incident occurs, an Administrator can flip the operator's status to **Deactivated**. Within sub-seconds, the remote operator's application halts operations, displays a security alert, and forcibly signs out to the login gate.

8.4 Immutable Audit Logging

All sensitive actions are permanently logged in the `audit_logs` table:

- Staff Created, Edited, Prorated, or Deactivated
- Salary Overrides and Monthly Variations Recorded
- Loan Facilities Disbursed, Paused, Resumed, or Manually Repaid
- Payroll Periods Locked, Finalized, or Exported
- Statutory Tax Parameters or CRA Formulas Modified

09 Troubleshooting, Data Backup & Disaster Recovery

Procedures for system maintenance, encrypted local backups, and resolving common operational edge-cases.

9.1 Local Encrypted Backups

Under **Settings -> Data Backup**, administrators can generate an encrypted `.json` database dump containing all company entities, employees, loans, locked historical runs, and audit logs.

> **Recommended Backup Schedule:** Export an encrypted local backup at the conclusion of every monthly payroll cycle immediately following period snapshot locking.

9.2 Disaster Recovery Procedure

1. Deploy a fresh Supabase database instance.
2. Execute `schema.sql` via the Supabase SQL Editor to initialize relational tables.
3. Launch Payroll Pro and enter the new Supabase URL and Anon Key at the login gate.
4. Navigate to **Settings -> Restore from Backup** and select your latest verified `.json` backup file.
5. The system validates and restores all companies, employees, and historical snapshots within seconds.

9.3 Frequently Asked Questions & Troubleshooting

SYMPTOM / QUESTION	ROOT CAUSE & RESOLUTION
Why does a staff member's take-home pay appear lower than expected?	Check the employee's profile for: (1) Mid-month Start or Exit date activating calendar day proration; (2) Scheduled loan/salary advance deduction; or (3) A monthly override value set for that cycle.
Table flickering or reloading constantly?	Resolved in version 1.0.2. Background synchronization and polling routines operate in silent mode, preserving table state and active row inputs without layout shifts.
Can I edit a staff member's salary after locking a period?	Yes. Updating base gross salary will update all future unlocked periods. The locked period snapshot remains permanently preserved as finalized.
How do I change statutory pension rates?	Administrators can navigate to **Settings -> Variables** to adjust Employee Pension (8%) and Employer Pension (10%) rates to match special collective bargaining agreements.